SmallRAG: When Compact Models with Retrieval Outperform Large Models

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Abstract

SmallRAG asks whether a compact retriever-generator pipeline can outperform a much larger closed-book system. A 3B Small Language Model distilled with TinyDistill is paired with RetGen-style retrieval. On Natural Questions, SmallRAG-3B beats a 70B closed-book model because retrieved passages supply facts the compact model cannot memorize. On HotpotQA, however, it falls behind when Multi-hop QA requires combining several pieces of evidence. The result is a clear trade-off: retrieval helps a small model know more, but does not automatically teach it to plan across multiple facts.

Keywords

Small models, Retrieval, Open-domain QA, Multi-hop QA

1. Introduction

Open-domain QA is often evaluated with large models, yet deployment may favor a Small Language Model. RetGen [1] provides the retrieval interface and TinyDistill [2] provides a compact generator. SmallRAG combines them and evaluates both direct questions and the harder Multi-hop QA setting, with a 70B closed-book model as a generic comparison.

2. Method

SmallRAG-3B starts from a TinyDistill checkpoint and uses RetGen to retrieve passages before generation. A lightweight question encoder selects the top evidence, which is concatenated with the question and passed to the Small Language Model. Training mixes answer supervision with a context-use loss that penalizes answers unsupported by retrieved text. The system has no external structured representation and uses the same inference budget on Natural Questions and HotpotQA. We report Exact Match and inspect whether the retrieved context contains all facts required for Multi-hop QA. This design keeps the model small while shifting part of the knowledge burden to retrieval.

3. Experiments

On Natural Questions, SmallRAG-3B reaches 58.2 EM versus 55.0 for the 70B closed-book model. On HotpotQA, it reaches 32.4 EM versus 35.7, confirming the multi-hop weakness. Compared with a 3B closed-book model, retrieval adds 8.6 points on Natural Questions and 4.1 points on HotpotQA, but the remaining gap is concentrated in questions requiring three or more evidence steps.

System	Natural Questions EM	HotpotQA EM
70B closed-book	55.0	35.7
SmallRAG-3B	58.2	32.4

4. Conclusion

SmallRAG shows that a Small Language Model with RetGen retrieval can outperform a larger closed-book model on direct Open-domain QA. Its failure on Multi-hop QA is the main open problem: retrieving facts is not enough when the system must connect them reliably.

References

[1] Vik, A. (2021). Retrieve-then-Generate: Augmenting Language Models with Retrieved Passages for Knowledge-Intensive Tasks. Conference on Natural Language Systems (CNLS).

[2] Tanaka, M. (2023). TinyDistill: Knowledge Distillation into Compact Language Models. Workshop on Efficient Language Models (WELM).